

# Usability Report

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# 1 Introduction

Evaluation of qualitative aspects of the Physio Companion app was done through a usability study. The project timeline and resources limit the participant pool size, and as a result, this study does not provide any quantitative insights. This document outlines the testing methodology and concludes with the resulting findings.

The goals of the study are:

- Assess the ease of use of the app across a diverse demographic
- User understanding of the terminology within the app
- User perception of the flow and navigation within the app
- Identification of any friction points
- Feedback on additional and beneficial features within the app

## 2 Techniques

The techniques used to conduct these tests were a mixture of both in-person and remote moderated testing. The mode employed was a questionnaire or an interview. This decision was up to the moderator. At the start of the study, participants were asked a series of questions inquiring some background information. The participants were then shown how to install the app on their smartphone. After successful installation of the program, the facilitator prompted the participant on the app's use. An additional investigator was playing the role of an observer recording the user's behaviour. At the end of the study, participants were asked for their final impressions. The study was voluntary and lasted approximately 30-40 minutes.

## 3 Methodology

In order to gain a deeper understanding of the user experience of the application, a series of questionnaires were constructed for each stage of testing ([pre-session](#), [session](#), and [post-session](#)). In addition, the 'session' questionnaire was further subdivided in order to examine the different main actions of the application. This breakdown is crucial to understanding the different facets of the application. Obtaining feedback promptly after completing each section will allow immediate and reliable reporting of the participants' impressions. The questionnaires were constructed as a guide and investigators were free to use either an interview-style approach or use the explicit questionnaire. If the former was used, it was essential that the topics mentioned in the questionnaire were addressed.

## 4 Participant profiles

The participant pool comprises of individuals of varying ages, genders and occupations, which ensures coverage of a broad target user base. The coverage details are outlined below:

- Name
- Age: Participants of various ages were chosen, with the sole restriction being that they had to be over 18.
- Gender: In an ideal case, an equal and all-encompassing balance of genders would be sought out. This would include 2 female or male identifying individuals and one non-binary individual.
- Person with disability: Another criteria that ensures diversity is enlisting participants with disabilities. This will aid in identifying any accessibility barriers that may go unnoticed. In an ideal case, at least 1 or 2 participants would be those who have a disability.
- Occupation: This criteria was selected to give insight into lifestyle factors. For the purposes of this study, any bias towards conducting it for the same occupation would skew the results.
- Level of digital literacy: This measure pertains to smartphone usage. It is divided into low, medium, or high expertise indicating how comfortable the user is operating the device.
- Familiarity with injury to the knee: This relates to how familiar the participant is with a knee injury, whether they are familiar with it through their occupation (e.g. orthopaedic surgeon, physiotherapist) or whether they have had a knee injury in the past.

### *Participant 1*

- Age: Late 20s
- Gender: Male
- Person with disability: No
- Occupation: Physiotherapist
- Level of digital literacy: High
- Familiarity with injury to the knee: Very familiar

### *Participant 2*

- Age: Late 20s
- Gender: Male
- Person with disability: No
- Occupation: Physiotherapist
- Level of digital literacy: High
- Familiarity with injury to the knee: Very familiar

*Participant 3*

- Age: early 60s
- Gender: Male
- Person with disability: No
- Occupation: Technician
- Level of digital literacy: High
- Familiarity with injury to the knee: Familiar

*Participant 4*

- Name
- Age
- Gender
- Person with disability
- Occupation
- Level of digital literacy
- Familiarity with injury to the knee

*Participant 5*

- Name
- Age
- Gender
- Person with disability

- Occupation
- Level of digital literacy
- Familiarity with injury to the knee

Due to project timeline constraints, the participant pool was limited in size, and posed challenges in recruiting individuals who satisfied the criteria mentioned above. Furthermore, it was not possible to recruit a participant with a knee injury. However, there were some that had undergone physiotherapy treatment for other health reasons.

Two of the participants were physiotherapists. They were recruited to assess the PT flow of the application and provide their perspective. The remaining participants varied in age and fulfilled different combinations of the criteria above.

## 5 Testing

All questionnaires' accompanying logic is explained [here](#).

### 5.1 Pre-session questionnaire

See [pre-session questionnaire](#).

### 5.2 Session questionnaire

See [session questionnaire](#).

### 5.3 Post-session questionnaire

See [post-session questionnaire](#).

## 6 Results

### Physiotherapists

The version of the test used referred to the PT flow of the application.

These were the main points in the tests that both of these participants mentioned.

- Both participants emphasized the inclusion of an RPE (Rate of Perceived Exertion) from the patient as an additional measure.
- They also suggested incorporated the number of sets performed in each exercise session on the Feedback tab.

## 7 Discussion

Usability testing proved to be an invaluable asset in evaluating the application. The results were enlightening and were immensely beneficial to the development team in refining the final product.

For the physiotherapist participants, the study was conducted in a conversational interview format, because it proved to be more natural than a questionnaire. The physiotherapists tested the PT flow of the application and highlighted some improvements (see above). They noted that the patient's subjective experience was an indispensable measurement, which is particularly valuable when determining exercise modifications. They provided examples of how this may indicate how difficult or easy the exercise is for the patient and whether their performance is impacted by other limitations (eg. pain). Including the number of sets is useful for the same reason. Both participants found the navigation and its use within the application to be straightforward. Their positive response to the application suggests that it has strong potential and can be advantageous to patient recovery.

## 8 Conclusion